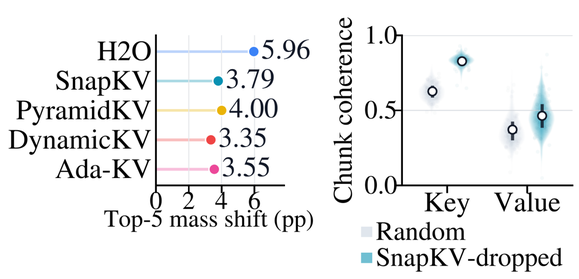
SurrogateKV: Representation-Preserving KV Cache Compression for Long-Context LLMs

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Motivation

Token eviction methods retain selected KV entries and remove the rest from at tention [1,7,8]. Some dropped regions cannot justify full raw storage, yet may s till contain information worth preserving in compact form.

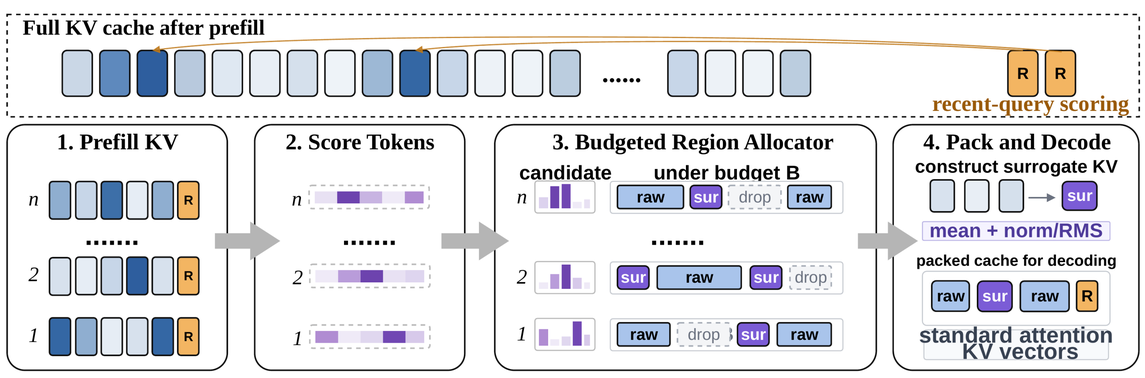


At B_{KV} = 512, eviction shifts top-5 ch unk attention mass by 3.35–5.96 pe rcentage points. Contiguous SnapK V-dropped chunks show higher key/ value coherence than matched ran dom sets.

SurrogateKV

After prefill, **SurrogateKV** allocates non-recent regions C_i under a fixed KV-slo t budget while keeping recent tokens R raw. High-salience regions retain RAW KV, selected lower-salience regions receive one independently addressable S URROGATE pair, and the rest are dropped.

$$|R| + \sum_{i:z_i=\text{RAW}} |C_i| + \sum_{i:z_i=\text{SURROGATE}} 1 \le B. \quad (7)$$



SnapKV [1], DynamicKV [2] and Ada-KV [3] supply shared, per-layer and per- head budgets. SurrogateKV groups contiguous tokens into 4-token atoms. Fro m recent-query attention, it ranks each atom by mean and peak scores separa tely. The higher rank sets raw retention, capturing high average attention or sh arp peaks.

Algorithm 1 SurrogateKV compression

Require: Q, K, V, budget B, recent size n_r
Ensure: Compressed cache $\widetilde{\text{KV}}$
Split cache into non-recent U and recent suffix R
Keep R raw
Scoring
$s_U \leftarrow$ recent-query token salience on U ▷ Eq. 3
$\{A_j\} \leftarrow$ contiguous atoms over U
$\{\phi_j\} \leftarrow$ atom salience from mean/peak ranks ▷ Eq. 5
Allocation
Initialize raw/drop atom actions under budget $B - |R|$
Seed candidates from dropped runs; merge across raw gaps when joint gain increases
Admit feasible candidates with $G_B(D) > 0$ ▷ Eq. 8
$\{(C_i, z_i)\}_{i=1}^{M} \leftarrow$ merge adjacent equal actions
Packing
for $i = 1, \ldots, M$ **do**
  if $z_i =$ SURROGATE **then**
    $\widetilde{\text{KV}}_i \leftarrow \{(\hat{k}_i, \hat{v}_i)\}$ ▷ Eqs. 10–11
  end if
end for
$\widetilde{\text{KV}} \leftarrow$ packed raw entries, surrogate pairs, and R
return $\widetilde{\text{KV}}$

Region allocation

Consecutive dropped atoms form candidate regions. Neighboring ca ndidates merge across raw atoms if joint gain exceeds their separate gains combined.

$$G_B(D) = \underbrace{v_{\text{S}}(D) + b_{\text{R}}(D)}_{\text{preserved value}} - \underbrace{\ell_{\text{in}}(D) + d_{\text{in}}(D) + \ell_{\text{pay}}(D)}_{\text{raw-evidence cost}} - \underbrace{c_{\text{reg}}[\Delta r(D)]_+ + p_B[\delta_B(D)]_+}_{\text{region and slot cost}}. \quad (8)$$

Gain balances preserved value ag ainst lost raw evidence and region /slot costs. Recompute it before a dmission and require positive gain within the budget.

Ablation

B_{KV} = 512, LLaMA-3-8B-Instruct.

Ablation	Setting	LB Avg.↑	# Sur.
Full method	SurrogateKV-Snap	**42.21**	397
No surrogate	Raw/drop only	40.44	0
Fixed quota	$\rho = 0.25$	40.51	253
	$\rho = 0.50$	40.33	505
	$\rho = 0.75$	40.40	757

Controls use only RAW/DROP or fix the surrogate count to test ada ptive selection. Adaptive admissio n scores 42.21, versus 40.44 with out surrogates and 40.33–40.51 with fixed quotas.

LB Avg. = LongBench average. # Sur. = mean surrogate count.

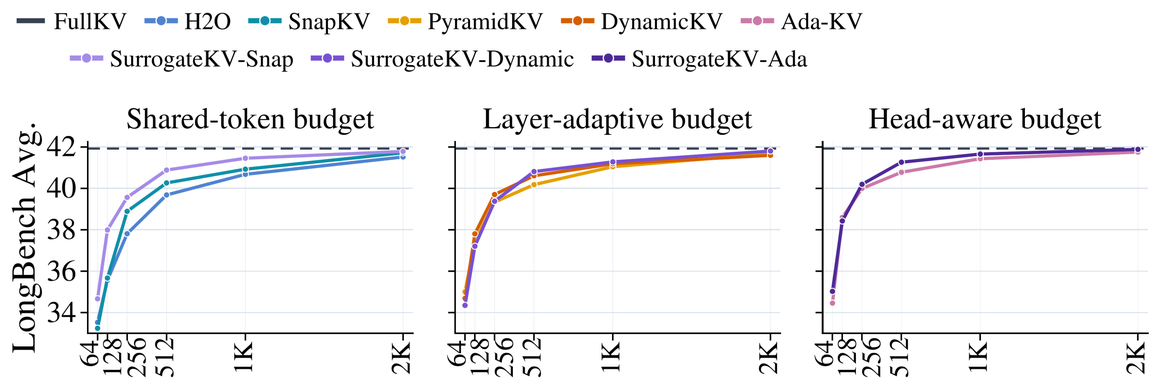
Conclusion

SurrogateKV preserves selected regions as independently addressable KV pa irs under a fixed budget. Selective admission improves LongBench quality and NIAH retrieval. Compression runs once after prefill and decoding uses standar d attention without retraining. This extends KV compression from token selecti on to representation allocation.

Evaluation

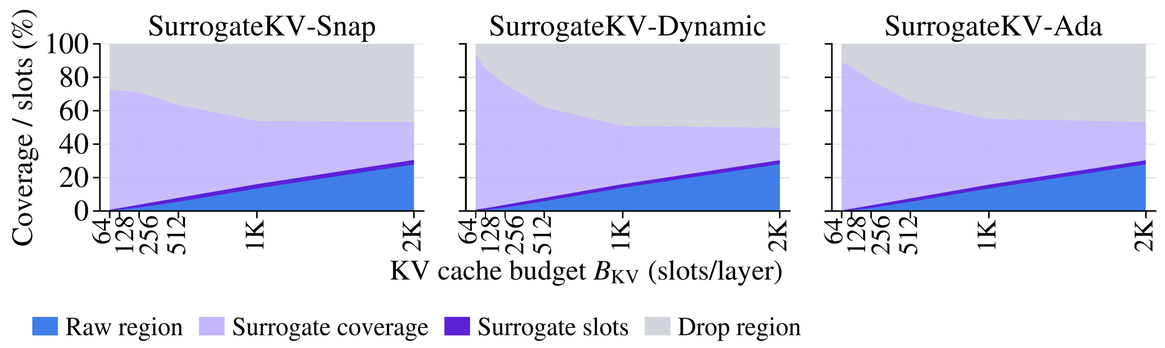
LongBench quality across budgets

LLaMA-3-8B-Instruct, 16 LongBench tasks [4], B_{KV} = 64–2048.



SurrogateKV-Snap outperforms SnapKV at all six budgets. SurrogateKV-Ada outperforms Ada-KV at five. At B_{KV} = 512, it scores 41.26 (Ada-KV: 40.77, FullK V: 41.92), retaining 98.4% of the full-cache score with 10.5% of its average inp ut KV cache.

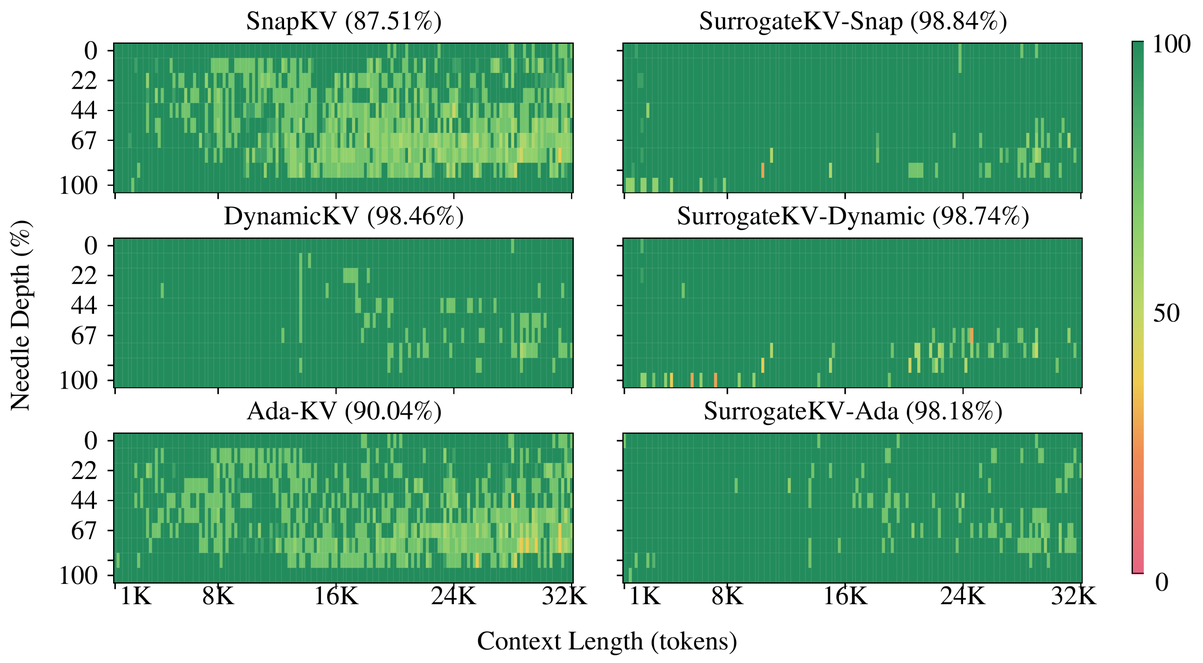
Allocation across cache budgets



One surrogate slot covers an entire region. Larger budgets retain more raw K V and shorten the spans represented by surrogates. The magnified dark band shows stored surrogate pairs.

Needle-in-a-Haystack (NIAH) retrieval

Retrieve a fact hidden at different positions in 1K–32K input tokens. Mistral-7B -Instruct-v0.2, B_{KV} = 128. Color shows accuracy (%) [9].



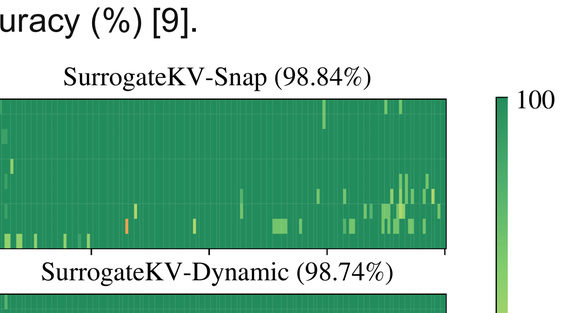
SurrogateKV improves all three baselines.
Mean gains over SnapKV and Ada-KV are 11.33 and 8.14 points, consistent with restoring coverage over evicted context.

Merging baselines

Method	LongBench score $B_{KV} = 128$	$B_{KV} = 512$
CaM	34.78	37.04
D2O	34.79	39.50
SurrogateKV-Snap	**37.99**	**40.88**

SurrogateKV-Snap leads CaM [5] and D 2O [6] at 128/512 slots. D2O uses appro ximate budgets.

First-token attention



B_{KV} = 64. Lower shift indicates concentration closer to FullKV.

Selected references

[1] Li et al., SnapKV, NeurIPS 2024. [2] Zhou et al., DynamicKV, Findings EMNLP 2025.
[3] Feng et al., Ada-KV, NeurIPS 2025. [4] Bai et al., LongBench, ACL 2024.
[5] Zhang et al., CaM, ICML 2024. [6] Wan et al., D2O, ICLR 2025.
[7] Zhang et al., H2O, NeurIPS 2023. [8] Cai et al., PyramidKV, COLM 2025.
[9] Kamradt, LLMTest Needle in a Haystack, GitHub 2023.

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Code & Paper